nature portfolio

Peer Review File

Experimental evidence of the topological obstruction in twisted bilayer graphene.

Corresponding Author: Dr Vincent Renard

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
This work studied quasiparticle interference (QPI) of twisted bilayer graphene through scanning tunneling microscopy and spectroscopy (STM/S). QPI signals arise from scattering at a naturally-formed defect reveal identical chirality of Dirac cones in the same valley, which is attributed to topological obstruction, as proposed in an earlier theoretical work (ref. 32).
The results are interesting, and the theoretical foundation is sound. However, substantial improvements are needed in the experimental section; therefore, I cannot recommend publication in its current form.
1. The main text and Supplementary Information indicate that the effective scatterer responsible for the observed QPI patterns is best described by a Gaussian potential, rather than a simple atomic vacancy or substitutional impurity. Given that the precise nature of the defect remains unidentified, can the authors present additional experimental evidence to support this modeling choice? It is helpful to have a systematic study on dI/dV spectra or spatial LDOS profiles acquired at and around the defect sites to further validate the Gaussian potential model over alternative scenarios. In particular, previous STM measurements on similar samples have reported defects due to irregularities in the interface of graphene and SiC, which had little scattering effects. Such possibility has to be considered.
2. The observation of QPI is closely linked to the presence and the density of point defects. Comparing the data in Fig. 1 and Fig. S5, it is somewhat counterintuitive that strong moiré-scale QPI is observed near a single defect in Fig. 1, whereas similar signatures are not evident in Fig. S5, despite a seemingly higher defect density. Could the authors clarify why the QPI signal is significantly stronger around the isolated defect in Fig. 1b? Additionally, how consistent and reproducible are the QPI features across different types of unintentional or engineered defects?
3. Many important contents are found in the supplementary materials. For clarity, the main text should specify which sections or figures are being referenced. Besides, some panels of Fig. 2 do not contain much useful information, whereas Fig. S3 appears to be more relevant. The authors may consider rearranging some panels in the revised manuscript.
4. As stated in the main text, this work serves as a complementary study to the earlier theoretical publication (ref. 32). The key finding of ref. 32 was the dislocation contrast of defects in AA and AB regions. However, the current study only reports QPI of a single defect (not in AA/AB) in a single device and does not discuss wave front dislocation. To strengthen the manuscript, I suggest that the authors measure additional defects/adatoms at different sites, as demonstrated in their previous publication.

Reviewer #2

(Remarks to the Author)
This paper provides an experimental investigation into the electronic properties of twisted bilayer graphene (TBG), with a particular focus on the phenomenon of "topological obstruction." Using STM, the authors examine the local density of states near defects in TBG and identify a distinctive pattern where Dirac cones within the moire minivalley share the same chirality. This observation provides strong evidence for the topological constraints predicted in the band structure of TBG. In addition, the authors confirm broader features of the band structure, such as Fermi velocity renormalization and a Lifshitz transition arising from Dirac cone mixing. These results are shown to be in good agreement with theoretical models. The paper is clearly written, and the subject is of high interest to the condensed matter and 2D materials communities. Provided that the concerns below are satisfactorily addressed, I would be inclined to support publication in Nature Communications, pending comments from other reviewers.

Comments

The manuscript states that moire-scale signals were not observed in another sample with similar twist angles. The absence of signal is attributed to the internal structure of defects suggesting that specific types of defects are necessary for enabling QPI measurements. However, this interpretation could be more rigorously supported by additional information on experimental reproducibility.

Was the visibility of the moire-scale signal sensitive to the STM tip condition?

Are there measurements taken with different setpoints that support the robustness of the observed features?

Can the authors provide evidence to rule out the possibility that the observed moire-scale signal is an artifact caused by the STM tip shape or other instrumental effects?

Clarifying these issues would strengthen the interpretation that the signal is intrinsic to the sample and defect structure, rather than a measurement artifact.

Reviewer #3

(Remarks to the Author)
The manuscript by Mesple et al. presents a comprehensive experimental theoretical investigation of the band structure in twisted bilayer graphene (TBG).Local density of states (LDOS) mapping around defects, combined with quasiparticle interference (QPI) analysis, reveals clear signatures of TBG′s topological properties. A four-band continuum description, together with tight-binding calculations, confirms the presence of topological obstruction and demonstrates that Dirac cones within a moir´e valley share the same chirality. At higher energies (above the van Hove singularity), the emergence of a star QPI pattern provides further evidence for a Lifshitz transition. The experimental data are of high quality, and the agreement between experiment and theory is compelling. I believe this manuscript meets the publication standards of Nature Communications, provided the following points are addressed.
Comments/Questions:
1. While the calculations predict robustness to different defect types and positions, do the experimental results confirm that QPI patterns around different defects provide evidence of topological obstruction? For example, have multiple defect sites been measured to verify this? Furthermore, could a complex-shaped defect, rather than a simple point defect, generate unconventional QPI pattern?
2. The manuscript does not explain why the scattering pattern can change from six-fold symmetry to an asymmetric form; please discuss possible physical mechanisms.
3. In Fig. 2a, explain the origin of the oscillatory features in the theoretical calculations.
4. In Fig. 3, I would suggest improving the readability of the figure- ensure that all symbols, colors, and lines are clearly defined. In particular, the green arrows and symbols are not explained. Clarify whether some plotted points are mirrored data lacking error bars, and if so, explain the reason.
5. Below the van Hove singularity, scattering patterns are also visible around the Γ point; please discuss their possible origin.
6. If applicable, please add citations to support the statement″For twist an gles below 10◦, the electronic structure of TBG supports a four-band con tinuum description in which the electrons behave as massless relativistic fermions with identical chirality in a given mini-valley″
7. Please discuss the presence or absence of intervalley backscattering be tween different mini-valleys in the measurements.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have addressed most of the comments and significantly improved the manuscript. I acknowledge that the “two-sublattice-contribution” theory matches the experimental data well and predicts site-independent results. However, a key concern, which is also noted by other referees, remains unresolved: QPI data were acquired from a single defect and therefore taken with a single microtip. Verifying site-independent QPI as well as the reproducibility by measuring more defects would substantially strengthen the manuscript, particularly for experimentalists. A full dataset across many bias voltages may not be necessary, given the difficulty of taking high-quality QPI.

I recommend acceptance pending the provision of additional data.

Reviewer #2

(Remarks to the Author)
The authors have provided satisfactory responses to my previous concerns. I have no further objections, and I recommend acceptance of the manuscript.

Reviewer #3

(Remarks to the Author)
The authors have addressed my previous concerns, and I consider the manuscript suitable for publication.

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The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.
To view a copy of this license, visit https://creativecommons.org/licenses/by/4.0/

REVIEWER COMMENTS

We thank the reviewers for their positive assessment of our work and their comments which helped to improve the manuscript. As a summary of the changes, we have reorganized the supplementary information so that the sections are ordered as they are cited in the main text. We have also added a clear reference to the supplementary figures and sections when needed. We have added a supplementary discussion concerning the determination of the type of defect from the comparison of the numerical model to experiments. We have also added an supplementary figure of the spatial dependence of the spectroscopy across the defect. We have modified Figure 2 and the legend of Fig. 3 to improve the readability. In the following the Figure numbers correspond to the ones of the revised manuscript.
We believe that the manuscript is now ready for publication.

Reviewer #1 (Remarks to the Author):

This work studied quasiparticle interference (QPI) of twisted bilayer graphene through scanning tunneling microscopy and spectroscopy (STM/S). QPI signals arise from scattering at a naturally-formed defect reveal identical chirality of Dirac cones in the same valley, which is attributed to topological obstruction, as proposed in an earlier theoretical work (ref. 32). The results are interesting, and the theoretical foundation is sound. However, substantial improvements are needed in the experimental section; therefore, I cannot recommend publication in its current form.

1. The main text and Supplementary Information indicate that the effective scatterer responsible for the observed QPI patterns is best described by a Gaussian potential, rather than a simple atomic vacancy or substitutional impurity. Given that the precise nature of the defect remains unidentified, can the authors present additional experimental evidence to support this modeling choice? It is helpful to have a systematic study on dI/dV spectra or spatial LDOS profiles acquired at and around the defect sites to further validate the Gaussian potential model over alternative scenarios. In particular, previous STM measurements on similar samples have reported defects due to irregularities in the interface of graphene and SiC, which had little scattering effects. Such possibility has to be considered.

In the previous version of the manuscript, we have provided in Fig. S5 a detailed comparison of the local density of states spectra near various types of defects. The remark of the referee reveals that we have not discussed these results deeply enough to convince the reader that the Gaussian potential is the best model of our defect. Let us now address this issue.

We would like to draw the attention of the referee on details of the experimental local density of states spectra that are only reproduced with a Gaussian potential. First, there is a bound state near the defect site and above the van Hove singularity at 0.65 eV (orange and blue curves in Fig S5c). This state is only reproduced by the Gaussian potential (defects d to l in Fig. S5). This bound state which appears just below the onset of the parabolic upper band is the expected consequence of an attractive potential in 2D system (E. Zaremba et al., Phys. Rev. Lett. 90, 046801 (2003), F.E. Olsson et al., Phys. Rev. Lett. 93, 206803 (2004)). None of the other types of defects (defects i to t in Fig S5) reflect this feature. Secondly, there is a marked increase of the local density of states in the range 0.1-0.35 eV near the defect (orange and blue curves in Fig S5c) which is also not reproduced by other type of defects. This relates to the enhanced QPI signal for this type of defect which microscopic origin is unclear at the moment but very well depicted by the model. Thirdly, there is a shift of the van Hove singularity to lower energies near the defect which is once again only describes correctly by the model with a Gaussian potential.

Together with the good agreement between simulated STM maps and experiment (Fig. S4), we believe that this detailed comparison establishes unambiguously the Gaussian potential as the best model of our defect. In addition, we note that we have adjusted the parameters of the Gaussian potential (depth and width) to have the best description of the experimental features (Fig. S5d)
We have included this discussion of Fig. S5 in the revised manuscript in supplementary section III .

In addition, as suggested by the referee, we have also included a more detailed spatial dependence of the local density of states near the defect in Figure S6 (Fig R1 below). This agrees well with the tight-binding calculations further strengthening our claim.

We believe that these additions improve the manuscript and we thank the referee for their comment.

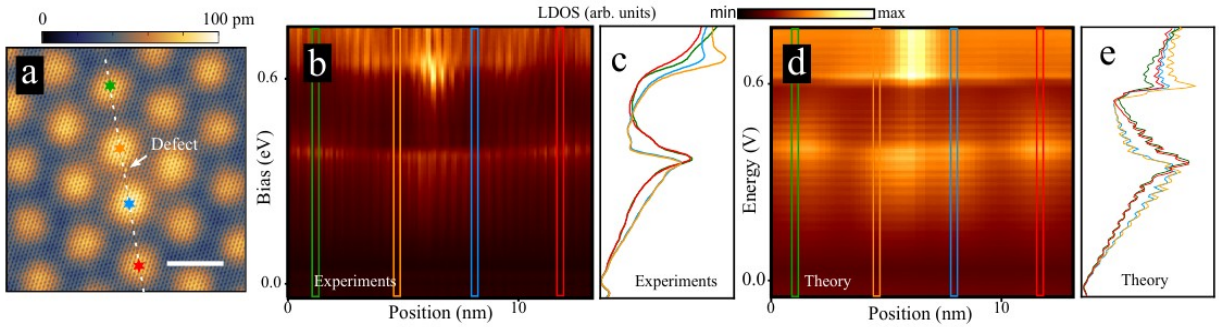

Fig R1 : line spectroscopy across the point defect. The short range oscillations in the experiment are due to top layer inter valley scattering QPI which are not discussed in the manuscript (see reply to question 7 of referee #3) and are not captured by the theory which has a lower spatial resolution (See supplementary section II)

2. The observation of QPI is closely linked to the presence and the density of point defects. Comparing the data in Fig. 1 and Fig. S5, it is somewhat counterintuitive that strong moiré-scale QPI is observed near a single defect in Fig. 1, whereas similar signatures are not evident in Fig. S5, despite a seemingly higher defect density. Could the authors clarify why the QPI signal is significantly stronger around the isolated defect in Fig. 1b? Additionally, how consistent and reproducible are the QPI features across different types of unintentional or engineered defects?

We would like to stress that not all defects give a sizable QPI signal. This is illustrated in the tight-binding calculations in Fig. S3. They show that the signal for a vacancy is an order of magnitude smaller than that for a Gaussian potential. The point of Fig. S8 (Formerly Fig S5) is precisely to demonstrate that experimentally. As it appears, only the defect discussed in the main text produce a measurable QPI signal while the defects present in Fig S8 do not produce appreciable signal at the scale of the moiré contrary to that of Fig. 1 of the main text. We were not able to find another defect similar to that discussed in the main text.

3. Many important contents are found in the supplementary materials. For clarity, the main text should specify which sections or figures are being referenced. Besides, some panels of Fig. 2 do not contain much useful information, whereas Fig. S3 appears to be more relevant. The authors may consider rearranging some panels in the revised manuscript.

We agree with the referee that labeling the specific supplementary section and Figures improves the readability of the paper. We have corrected this in the present manuscript.

Figure 2 is essential to our demonstration : There are arcs in the Fourier transform of the local density of states independently of the position and nature of the defect so that our claim is very robust. As a consequence, we believe Figure 2 should remain as is (See also reply to comment #4).

Still, we agree with the referee that the previous version of the figure was difficult to read. Therefore we added labels to ease the reading and simplify its legend.

Concerning Figure S4 (Formerly Fig. S3), we would like to stress that the information displayed in it is redundant with that of Figure 2 and 3 of the main manuscript. We included Fig S4 for sake of completeness in the direct comparison of experiments and theory.

4. As stated in the main text, this work serves as a complementary study to the earlier theoretical publication (ref. 32). The key finding of ref. 32 was the dislocation contrast of defects in AA and AB regions. However, the current study only reports QPI of a single defect (not in AA/AB) in a single device and does not discuss wave front dislocation. To strengthen the manuscript, I suggest that the authors measure additional defects/adatoms at different sites, as demonstrated in their previous publication.

Ref.32 is a strong motivation to our present study as it proposes to exploit QPIs at the moiré scale to reveal the topological obstruction. However, there are two fundamental differences between Ref.32 and our work, implying that the predictions in Ref. 32 do not apply to our work.

1- The predictions of Ref.32 are based on the assumption that the tip couples equally to the upper and lower layers, such that the predicted signal results from equal contributions of the four sublattices. In contrast, the STM tip mainly probes the LDOS of the upper layer in our experiment and the predicted signal results from the contributions of the two upper sublattices. The LDOS signals predicted in Ref.32 and our work are therefore different and are not expected to lead to the same QPI patterns.
In particular, the four-sublattice-contribution description in Ref.32 leads to complete 2q-circles in the QPI Fourier space for all defect sites (see Supplemental Materials in Ref. 32). In contrast, our two-sublattice-contribution description predicts single extinctions in the 2q-circles independently of the defect site, which is also in agreement with the experiment.

2- The information on the wavefront dislocations in real space in Ref. 32 is captured equivalently in the QPIs (modulus+phase) in Fourier space. For instance, they manifest as phase vortices in the intervalley scattering vectors in Fourier space [Nature 574, 219 (2019)]. We can therefore study the signal in either space. While the dislocations in Ref.32 involve the knowledge of both the QPI modulus and phase in Fourier space and depend on the defect location (AA, AB, etc), our prediction relies on the QPI modulus only and is independent of the defect location. Once again, our prediction is intrinsically different from the wavefront dislocations predicted in Ref. 32.

Thus, we believe that the good agreement with the experiment (point 1) and the fact that it does not involve the QPI phase and defect position (point 2) make our prediction a more robust manifestation of the topological obstruction than the wavefront dislocations, where we do not control the nature/position of the defect and do not measure a significant contribution of the bottom layer. We agree with the referee that a careful investigation of the phase of QPIs is also interesting but this is beyond the scope of the present study.

We would like to mention that Dario Bercioux presented a similar theoretical calculation at the graphene conference 2025 which independently confirms our result. (D. Bercioux, Chiral Interlayer Quasi-Particle Interference in Twisted Bilayer Graphene. Graphene 2025, Donostia Spain). Here is a link to the abstract.

Reviewer #2 (Remarks to the Author):

This paper provides an experimental investigation into the electronic properties of twisted bilayer graphene (TBG), with a particular focus on the phenomenon of "topological obstruction." Using STM, the authors examine the local density of states near defects in TBG and identify a distinctive pattern where Dirac cones within the moire minivalley share the same chirality. This observation provides strong evidence for the topological constraints predicted in the band structure of TBG. In addition, the authors confirm broader features of the band structure, such as Fermi velocity renormalization and a Lifshitz transition arising from Dirac cone mixing. These results are shown to be in good agreement with theoretical models. The paper is clearly written, and the subject is of high interest to the condensed matter and 2D materials communities. Provided that the concerns below are satisfactorily addressed, I would be inclined to support publication in Nature Communications, pending comments from other reviewers.

Comments

The manuscript states that moire-scale signals were not observed in another sample with similar twist angles. The absence of signal is attributed to the internal structure of defects suggesting that specific types of defects are necessary for enabling QPI measurements. However, this interpretation could be more rigorously supported by additional information on experimental reproducibility. Was the visibility of the moire-scale signal sensitive to the STM tip condition?

The tip condition is indeed an important parameter, and STM tips can degrade while scanning over long periods of time, in particular when scanning is done close to the surface. In order to mitigate this issue, we have performed all our DOS maps at a constant tunneling resistance of 2 GΩ ensuring minimal tip/sample interaction, allowing us to record all the relevant set of data with a single microtip.

The following figure shows that the QPI signal is insensitive to the tunneling conditions.

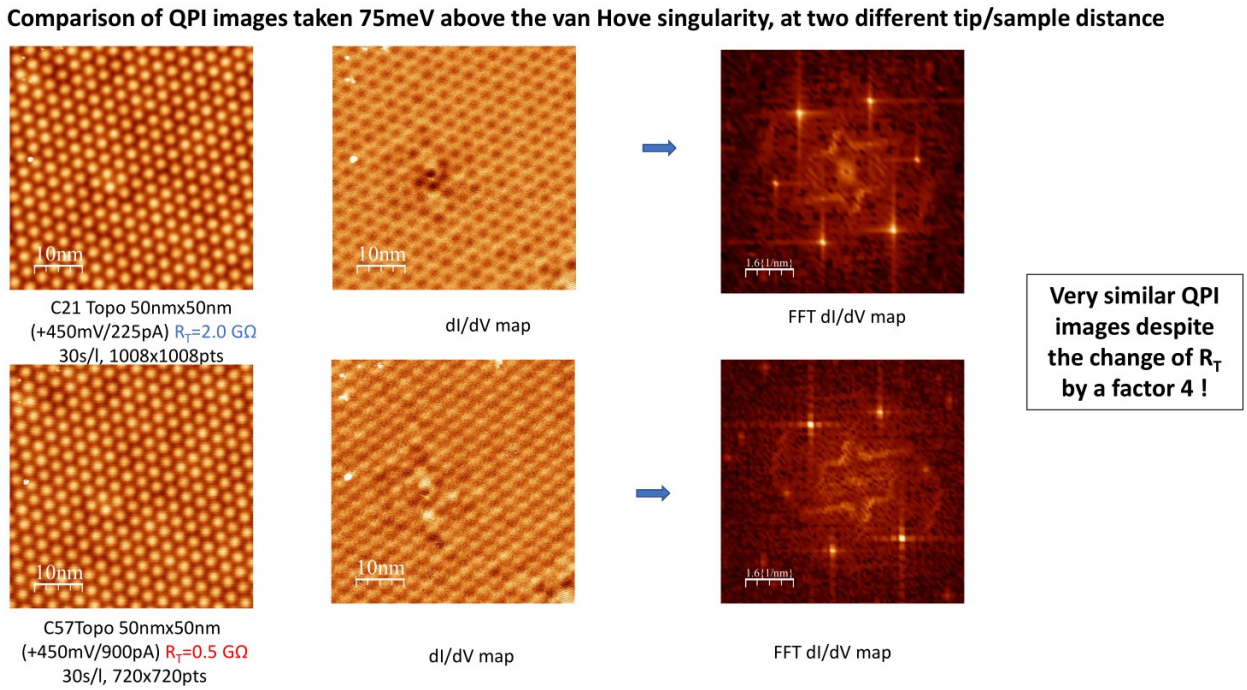

Figure R2 : Robustness of the measurements to tunneling conditions.

At the end of our measurement session we did observe a tip change and a general degradation of the surface. Yet the chiral QPI pattern remained visible.

Are measurements taken with different setpoints that support the robustness of the observed features?

In our experience, the set point does not affect the robustness of the observed features, although it can increase the visibility by increasing the LDOS signal, as shown on Figure R3 below. As mentioned above, we have focused on keeping a constant tip height throughout the recording of the experimental images.
The figure below illustrates that the spectroscopy is unaffected by a ten times change of the tunneling resistance

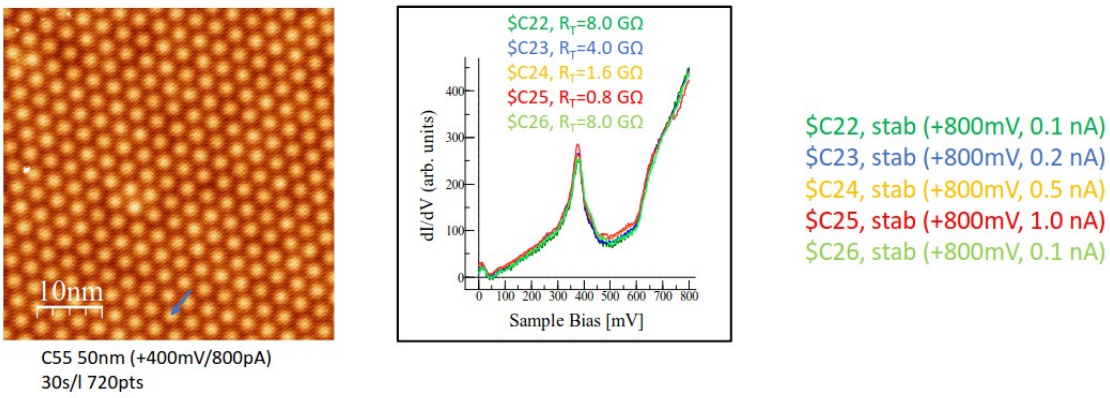

Figure R3 : STS spectra for various tunneling conditions on the sample shown on figure S5, showing no changes in the relevant features of the spectra.

Can the authors provide evidence to rule out the possibility that the observed moire-scale signal is an artifact caused by the STM tip shape or other instrumental effects?

The usual artifacts in an STM measurement are multiple tips in which a feature is imaged multiple times, deformations related to thermal drift or a bad calibration of the piezzo scanner. We do not see any indication of such artifacts in our images. In addition, since the pattern we observe has a well defined origin at the defect, the defect must be the cause of the pattern. In addition, we achieve a very good description of the image with the tight-binding calculations which do not include the STM tip at all. We believe that this is the most important evidence that we are indeed probing the physics of this defect excluding any experimental artifact.

Reviewer #3 (Remarks to the Author):

The manuscript by Mesple et al. presents a comprehensive experimental theoretical investigation of the band structure in twisted bilayer graphene (TBG). Local density of states (LDOS) mapping around defects, combined with quasiparticle interference (QPI) analysis, reveals clear signatures of TBG′s topological properties. A four-band continuum description, together with tight-binding calculations, confirms the presence of topological obstruction and demonstrates that Dirac cones within a moiré valley share the same chirality. At higher energies (above the van Hove singularity), the emergence of a star QPI pattern provides further evidence for a Lifshitz transition. The experimental data are of high quality, and the agreement between experiment and theory is compelling. I believe this manuscript meets the publication standards of Nature Communications, provided the following points are addressed.

1. While the calculations predict robustness to different defect types and positions, do the experimental results confirm that QPI patterns around different defects provide evidence of topological obstruction? For example, have multiple defect sites been measured to verify this? Furthermore, could a complex-shaped defect, rather than a simple point defect, generate unconventional QPI pattern?

We have measured many defects inducing intralayer scattering such as those from figure S8c-d-e (Previously Fig. S5). Unfortunately, among this zoo of defects only the one presented in the main text shows the specific QPI pattern. We believe that the excellent agreement with the analytical and numerical model is sufficient to support our claim. Equation 1 of the main text shows that the intensity prefactor is universal in that it relates to the wave functions of the bare material without the defect and does not depend on the specific strength/location of the defect. In this sense the observed 2q-arcs are the most simple expression of the topology of the wavefunctions of twisted graphene layers. The defect contributes to the T-matrix term in Eq. 1 and can indeed alter the pattern. It can for example reduce the amplitude as for the defects reported in Fig. S3.

2. The manuscript does not explain why the scattering pattern can change from six-fold symmetry to an asymmetric form; please discuss possible physical mechanisms.

There are several effects affecting the symmetry of the signal. At low energies, the main effect is the position of the defect within the moiré unit cell. Away from the center of AA regions the QPI signal is weakened in a direction (Fig 2). At present we have no explanation for this effect which is yet very well captured by the tight-binding model. This is seen even in absence of strain. At the van hove singularity and above, heterostrain has more influence. This can be seen in the calculated band structure shown below. Please look at the constant energy contours near the Van Hove singularity in orange where the six fold symmetry is affected.
We have added a comment in the main text but the detailed description will deserve a more complete theory beyond the present study.

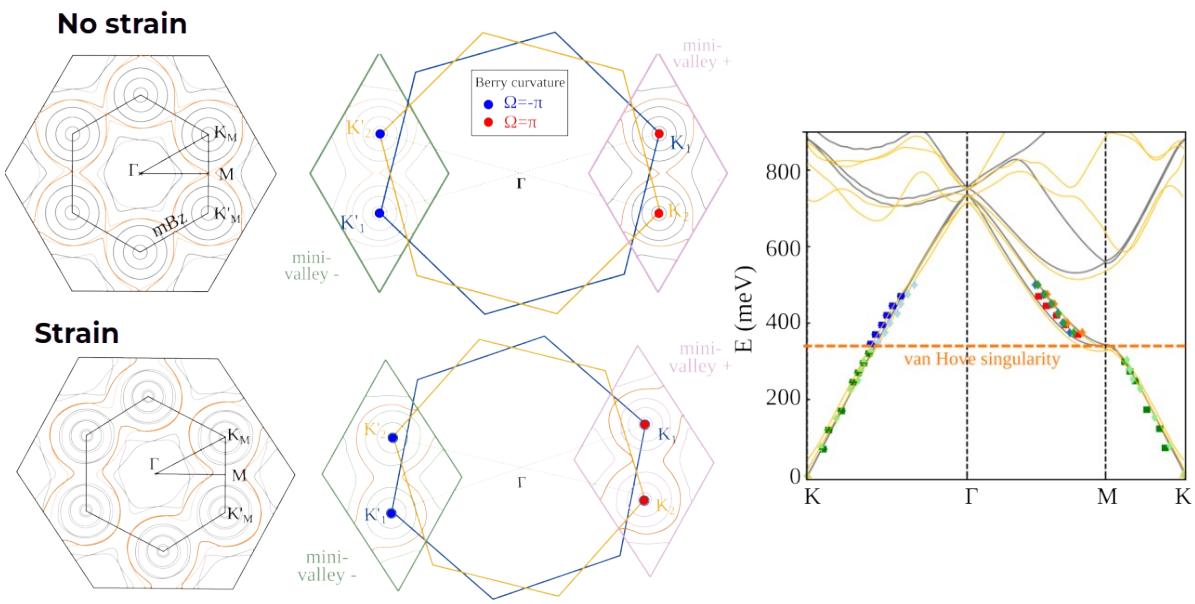

Figure R4 : effect of heterostrain on the and structure.

3. In Fig. 2a, explain the origin of the oscillatory features in the theoretical calculations.

These oscillations are a well controlled artifact of the calculation coming from size effects in the calculated cell (D. G. Pettifor, D. L. Weaire (Eds), The recursion method and its applications, Springer Series in Solid-State Sciences 58 (Springer, Berlin, Heidelberg,

1987). https://doi.org/10.1007/978-3-642-82444-9). These are oscillations are mitigated by a convolution with a Gaussian of standard deviation 5 meV.We have clarified this in the revised manuscript supplements.

4. In Fig. 3, I would suggest improving the readability of the figure- ensure that all symbols, colors, and lines are clearly defined. In particular, the green arrows and symbols are not explained. Clarify whether some plotted points are mirrored data lacking error bars, and if so, explain the reason.

We thank the referee for pointing the issue of the readability of Figure 3. We have clarified the arrows and symbols in the caption of Fig 3. Yes indeed, at low energy (below the van Hove singularity) the data along KΓ are replicated along KM for illustration purpose. This is justified because the small strain does not lead to any visible deformation od the 2q-rings as expected.
If the error bars were not visible it is because they were too small compared to the symbol size. This is now corrected

5. Below the van Hove singularity, scattering patterns are also visible around the Γ point; please discuss their possible origin.

The experimental QPI signal around the Γ point shows that there are intravalley scattering processes. Intravalley scattering can occur in four non-equivalent valleys: K1+, K1-, K2+, and K2-. Here again, we expect from a stationary phase approximation (see Sec. I.C in SI) that back-scattering is the dominant process. However, we may expect that intravalley back-scattering is forbidden in the top layer (contributions from K1+ and K1-), since the top-sublattice components of the wave-functions in Eq.(S1) are identical to that in graphene, responsible for the absence of intravalley back-scattering. This may then reduce the signal to Friedel oscillations decaying as 1/r^2 (instead of 1/r). In addition, there is also the contribution from the K2+ and K2-, for which we expect back-scattering given the top-sublattice components of the wave-functions in Eq.(S2). However, this contribution should be small as the wavefunctions near K2 in the top plane are attenuated by a factor ten compared to that at K1. These are the reasons why, apart from pure back-scattering, we expect some intravalley-scattering signal around the Γ point.

Note that this signal does not depend on the distribution of the valleys in reciprocal space and so does not tell whether the Dirac cones have the same or opposite chiralities in the mini-valleys, in contrast to intervalley scattering investigated in our work. As a consequence we do not discuss this effect in our manuscript.

6. If applicable, please add citations to support the statement "For twist angles below 10◦, the electronic structure of TBG supports a four-band continuum description in which the electrons behave as massless relativistic fermions with identical chirality in a given mini-valley"

This refers to the theoretical description made in ref. 32. We have added the reference in the manuscript.

7. Please discuss the presence or absence of intervalley backscattering between different mini-valleys in the measurements.

This is indeed an interesting question and we thank the referee for raising this point. The defects of Fig S8 do indeed induce inter-valley scattering in the graphene top layer as explained in the corresponding discussion. More interestingly, the defect of Fig. 1 of the main manuscript also induces a complex QPI pattern at the K points of the top graphene layer (see the figure below). This signal can be ascribed to scattering between minivalleys. As can be seen from the image, however, the signal is quite complex and would necessitate an careful study of it own and we leave it to future studies.

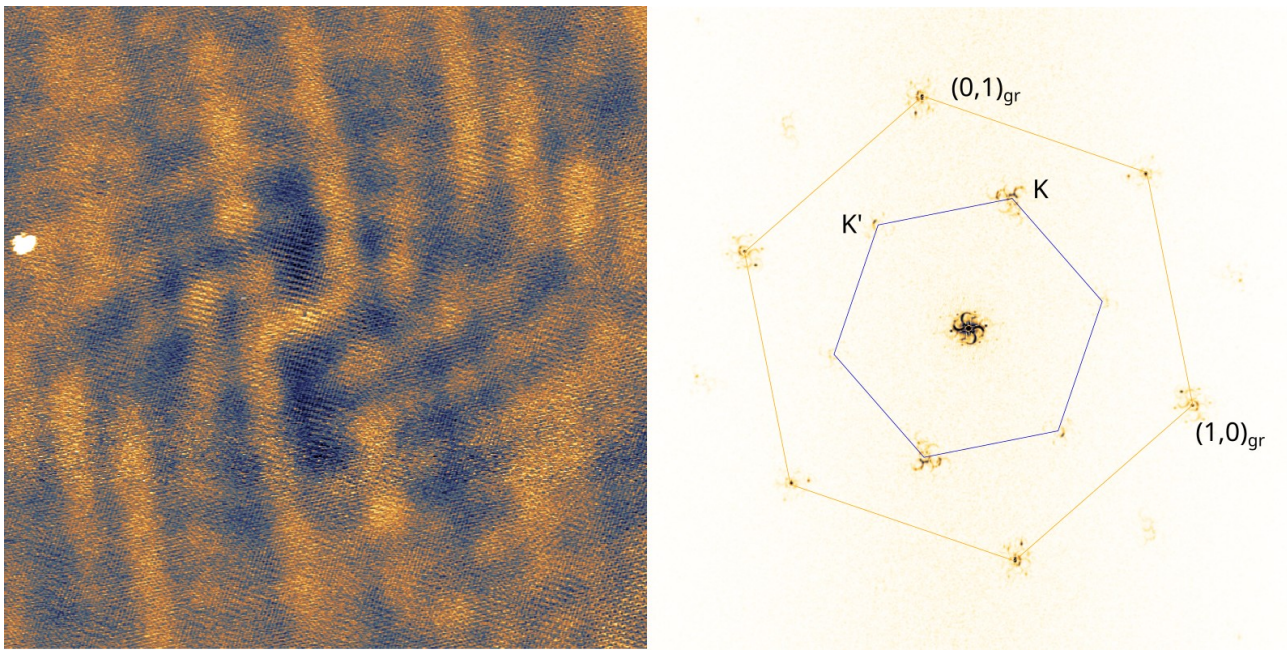

Figure R5: LDOS image measured at the energy E_{VHS}-100 meV and it Fourier transform. A complex QPI signal is seen at the K points of the top layer indicating inter-minivalley scattering.

Grenoble October 14th 2025

Dear Editor,
We thank the reviewers for their positive reviews of our results and the editor for the acceptance in principle of the manuscript for publication.
The final version of the manuscript was updated to remove the citations from the abstract and include sections and subsections in the main text according to the formatting guidelines. Other instructions were also respected (see the authotr checklist). Below we provide a response to the reveiwers last comments (responses are in blue).
Sincerely yours,

The authors

Reviewer #1 (Remarks to the Author):

The authors have addressed most of the comments and significantly improved the manuscript. I acknowledge that the “two-sublattice-contribution” theory matches the experimental data well and predicts site-independent results. However, a key concern, which is also noted by other referees, remains unresolved: QPI data were acquired from a single defect and therefore taken with a single microtip. Verifying site-independent QPI as well as the reproducibility by measuring more defects would substantially strengthen the manuscript, particularly for experimentalists. A full dataset across many bias voltages may not be necessary, given the difficulty of taking high-quality QPI.

I recommend acceptance pending the provision of additional data.

We thank the reviewer for their positive assesment of our work. In our previous reply, we have already adressed the concerns of the reviewer. We explained that the defect had also been measured after a tip change (i.e with a different microtip) without any qualitative change. Furthermore, we discussed our measurement of many more defect and the reasons why they do not all show the same QPI response. Only one of them showed the interesting QPI signal at the moiré scale which we attributed to the details of this defect.

Reviewer #2 (Remarks to the Author):

The authors have provided satisfactory responses to my previous concerns. I have no further objections, and I recommend acceptance of the manuscript.

Reviewer #3 (Remarks to the Author):

The authors have addressed my previous concerns, and I consider the manuscript suitable for publication.